Southern Cross University

Pharmacology of the Cardiovascular System

Hypertension

Ken Wojcikowski



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Pharmacology of the Cardiovascular System: Hypertension

COMMONWEALTH OF AUSTRALIA

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Medications of the Cardiovascular System:

Antihypertensive Medications

Lesson 1:

Introduction; Physical factors involved in blood pressure

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Objective

Explain the regulation of blood pressure, including the short term regulation and the long term regulation.

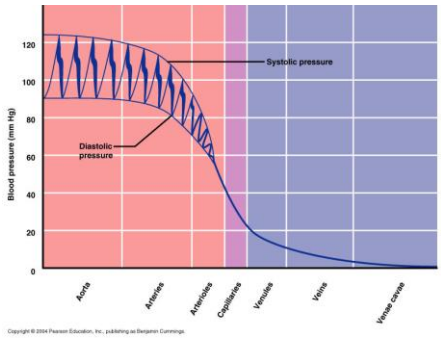
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Arterial Blood Pressure

- **Systolic pressure** - arterial blood pressure generated during **ventricular contraction**
- **Diastolic pressure** - arterial blood pressure generated when the **ventricles are relaxed**

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Factors Influencing Mean Arterial Blood Pressure

- **Physical**
  - Blood volume and the elastic properties of the blood vessels and narrowing of the blood vessels
    - Therefore, pathological factors like atherosclerosis can influence blood pressure
- **Physiologic factors**
  - Cardiac output (stroke volume x heart rate)
  - Peripheral vascular resistance
  - (Blood pressure = Cardiac output x peripheral vascular resistance)

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Medications of the Cardiovascular System:
Antihypertensive Medications
Lesson 2:
Physiological factors involved in blood pressure regulation

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Physiological Factors Influencing Blood Pressure

- Blood pressure = cardiac output × peripheral vascular resistance (total peripheral resistance)
  - CO
    - stroke volume x heart rate
      - sympathetic nervous system is most important
  - PVR is dependent on three factors:
    - Length of the vessel
    - Viscosity of the fluid
    - But mainly controlled by the diameter of the arterioles
      - Sympathetic nervous system is most important

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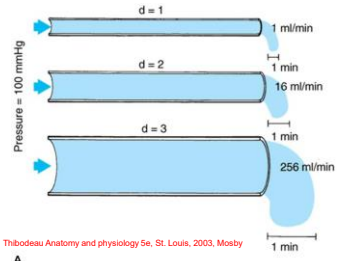
Resistance

Total Peripheral resistance (TPR)

- Contributing factors –
  - viscosity,
  - length of vessel (obesity)
- But resistance depends mostly **on radius of the vessel**
- The radius of the **arterioles** is most important
  - narrowing arterioles decrease blood pressure downstream and increase blood pressure upstream

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If a vessel lumen is decreased by half, it results in 1/16th the flow!



Thibodeau Anatomy and physiology 5e, St. Louis, 2003, Mosby

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Control of Arteriole Diameter

Arteriole diameter is most important in influencing resistance

- **Controllable factors that can affect arteriole diameter (resistance) and therefore affect blood flow and blood pressure include**
  1. **Neural mechanisms**
  2. **Hormonal stimulation**
  3. **Local effects**

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Review: most important factors for blood pressure regulation

- Heart rate
  - Mostly influenced by ANS
- Stroke volume
  - Volume of fluid
    - ADH the water saver, aldosterone the salt saver and ANP the salt waster)
  - And stroke volume
    - ANS
- Diameter of the arterioles
  1. Neural mechanisms
     - ANS
  2. Hormonal stimulation
     - ANS, angiotensin
  3. Local effects
     - Nitric oxide, prostaglandins, other mediators

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Medications of the Cardiovascular System:

Antihypertensive Medications

Lesson 3:

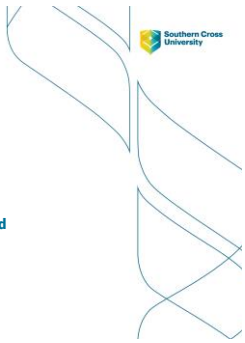
Physiological factors involved in blood pressure regulation (cont.)

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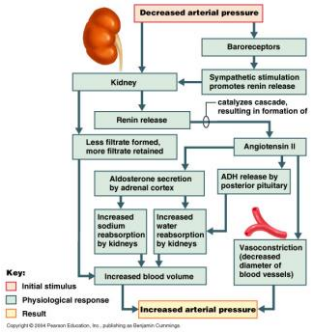
Hormonal stimulation

Renin angiotensin aldosterone mechanism

- Any decrease in the blood flow to the kidneys results in release of Renin
- Think of renin as a stress hormone released to save us from dehydration or blood loss



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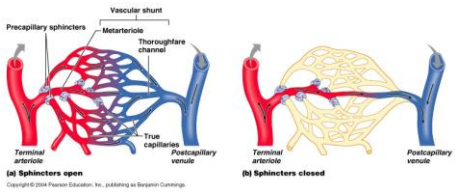
Control of blood pressure: Local effects

- Blood vessels have mechanisms that regulate their own blood flow
  - "Autoregulation"
    - Allows some organs to maintain constant blood flow between wide range of BP's
- Tissue factors contributing to local control of blood flow
  - Nitric oxide; Histamine
    - Increases blood flow
    - Provides autoregulation in skeletal muscle



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Local effects often work on precapillary sphincters



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Control of blood pressure: Local effects

The prostaglandins

- Some cause dilation
  - Prostaglandin E2
- Others cause constriction
  - Prostaglandin TXA2

Example:

The vasoconstriction of TXA2 from platelets...
...are balanced by the vasodilation effects of endothelial- derived vasodilator, Prostacyclin I 2

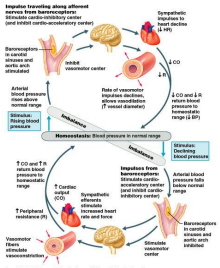
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Mechanisms of Blood Pressure Regulation

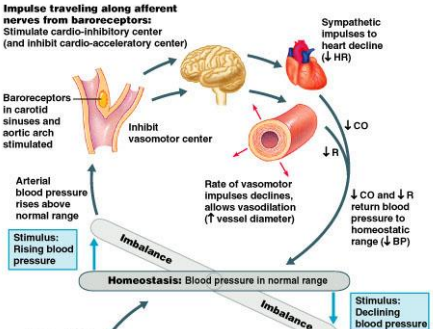
- **Short-term regulation:** corrects temporary imbalances in blood pressure
  - Neural mechanisms
  - Hormonal mechanisms
- **Long-term regulation:** controls the daily, weekly, and monthly regulation of blood pressure
  - Renal mechanism

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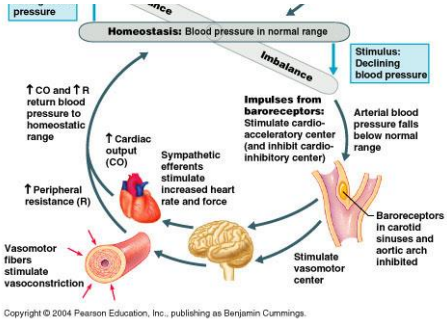
Summary of blood pressure regulation (see next two slides for enlarged version)



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Pharmacology of the Cardiovascular System

Medications for Hypertension

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Objective

Identify and describe the therapeutic actions and major side effects of common pharmacological agents used in the treatment of hypertension

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Medications of the Cardiovascular System:
Antihypertensive Medications
Lesson 4:
Overview of hypertension

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Overview of Hypertension



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Hypertension

A consistently elevated blood pressure equal to or greater than 140/90mmHg

Primary/essential hypertension (90-95% of cases)
- Many causes:
  - Lifestyle factors
  - Genetics

Secondary hypertension
- One cause
  - Underlying disease process
  - Drugs
  - Hormonal factors

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Medications of the Cardiovascular System:
Antihypertensive Medications

Overview of antihypertensive agents

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Medications to treat hypertension

Treatment of hypertension often involves several drugs

To help remember them, remember your A, B, C, D's

- A
- B
- C
- D

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Medications to treat hypertension

- Alpha 1 BLOCKERS (…zosin drugs)
- Alpha 2 AGONISTS (not antagonists; Methyldopa (for pregnancy)
- Angiotensin converting enzyme inhibitors (…pril drugs)
- Angiotensin 2 receptor blockers (ARBS; …sartan)
- Beta blockers (…olol drugs)
- Calcium channel blockers (Dihydropyridines)
- Diuretics – Especially Hydrochlorothiazide or thiazide diuretics

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Medications of the Cardiovascular System:
Antihypertensive Medications
Lesson 5:
Alpha 1 antagonists

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Alpha antagonists

Prazosin, doxazosin and other …zosin drugs

Alpha antagonists are also known as selective sympatholytic agents, alpha receptor blockers

- Block alpha 1 receptor sites on blood vessels
  - Decreased sympathetic stimulation to blood vessels
  - Decreased vascular tone
  - Vasodilation
  - Decreased vascular resistance
  - Decreased BP



Location of receptors:
- Blood vessels
- Prostate
- Urinary sphincters

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Alpha antagonists

Prazosin, doxazosin and other …zosin drugs

Alpha antagonists are no longer the preferred medication for hypertension or benign prostatic hyperplasia.

They may be best placed in people who have both BPH and hypertension at the same time.



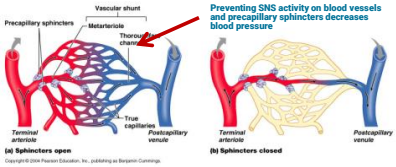
Location of receptors:
- **Blood vessels**
- **Prostate**
- **Urinary sphincters**

https://www.telegraph.co.uk/news/newstopics/howaboutit.ht

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Alpha antagonists

- **Block the alpha receptors located on arterioles and pre-capillary sphincters so that the SNS cannot**
  - **Lowers systemic vascular resistance.**

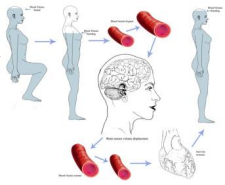


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Alpha antagonists

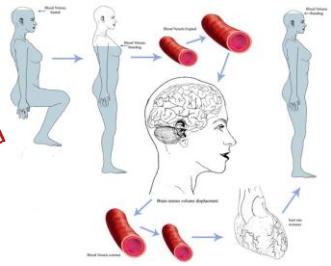
Alpha antagonists are also known as selective sympatholytic agents, alpha receptor blockers

- **Block receptors on prostate/ urinary sphincters**
  - **Potentially useful in Benign prostatic hyperplasia**
- **Adverse effects:**
  - **ORTHOSTATIC HYPOTENSION!!** →
  - **Syncope**
  - **Dizziness**
- **Prazosin (Minipress)**



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Orthostatic (postural) hypotension



Important safety issue! Teach client how to get up from a seated position safely!

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Medications of the Cardiovascular System:
Antihypertensive Medications

Alpha 2 agonists (Methyldopa for use during pregnancy: see end of lesson 8)

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Medications of the Cardiovascular System:
Antihypertensive Medications
Lesson 6:
Angiotensin system medications:
Medications of the Renin Cascade

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Medications of the Renin Cascade

Angiotensin-converting enzyme (ACE) inhibitors
…PRIL DRUGS including Ramipril

Angiotensin II receptor blockers (ARBS)
…SARTAN drugs like Candesartan

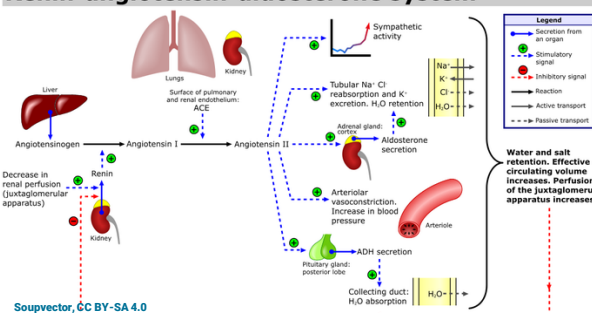
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Understanding the Renin Cascade

- **Think of renin as the panic button to simultaneously increase blood pressure by every means available**
- **When our blood pressure is too low, or there is sympathetic nervous system stimulation to the kidneys, renin is released.**
- **The ultimate effect is an increase of the blood pressure by at least 5 mechanisms**
- **Our medications of the renin cascade attempt to stop this cascade at two different points**

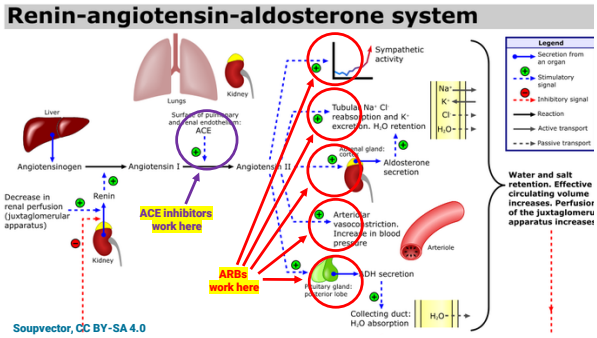
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Renin-angiotensin-aldosterone system



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Medications of the Renin Cascade

Renin-angiotensin aldosterone system

- **ACE Inhibitors inhibit the conversion of angiotensin I to angiotensin II**
- **ARBs antagonise the Angiotensin II receptors**

- **Both:**
  - **Decrease aldosterone the salt saver**
  - **Decrease ADH, the water saver**
  - **Decreases vasoconstriction**
  - **Increase fluid released from kidneys**
  - **Decrease the renin-controlled SNS 'positive feedback' loop**

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Angiotensin converting enzyme inhibitors

- **ACE inhibitors inhibit the conversion of angiotensin I to angiotensin II**
- **Ramipril and other …PRIL drugs**
- **Indicated in congestive heart disease and hypertension**
- **Adverse effects:**
  - **Chronic dry cough,**
  - **Hyperkalaemia**
- **Contraindicated in pregnancy (Category D)**

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Angiotensin II receptor blockers

Bind to angiotensin II receptors blocking the effect

- Candesartan and other …SARTAN drugs
- Indicated in Congestive Heart Disease and hypertension and PRE-HYPERTENSION!
- Adverse effects:
  - Fewer than ACE inhibitors
  - GI disturbances
- Contraindicated in pregnancy
  - Pregnancy category D



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Medications of the Cardiovascular System:
Antihypertensive Medications
Lesson 7:
Beta blockers

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Beta blockers

Beta blockers are sympathetic antagonists, sympatholytics, beta receptor blockers

- Example: Atenolol and other …OLOL drugs
- Block Beta 1 receptor sites on the heart
- Decreased sympathetic stimulation to the heart
- Decreased heart rate (negative chronotropic effect)
- Decreased contractility (negative inotropic effect)
- There are at least 4 drug classes that are better!
- Taper these drugs off slowly or a rebound hypertension may develop



Location of receptors:
- Heart
- Kidney (renin release)

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Beta blockers – third generation

- Carvedilol is considered a third generation beta blocker but does not have the …olol suffix due to having alpha 1 blocking properties.
- Indicated in high blood pressure, congestive heart failure (CHF) but not the first line therapy, due to side effects in over 10% of individuals taking carvedilol:
- Dizziness, Fatigue, Low blood pressure
- Diarrhea
- Weakness
- Bradycardia
- Weight gain
- Erectile dysfunction



Location of receptors:
- Heart AND
- Blood vessels

https://www.telegraph.co.uk/news/newstopics/howaboutthat

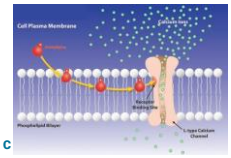
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Medications of the Cardiovascular System:
Antihypertensive Medications
Lesson 8:
Calcium channel blockers

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Calcium channel blockers

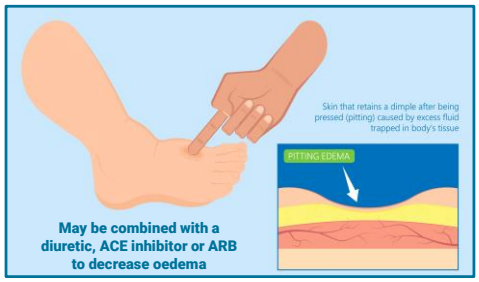
- Reduction in calcium inside the cells
  - Reduced cardiac contractility
  - Vasodilation, which reduces myocardial workload
- Adverse effects:
  - Peripheral oedema
  - Dizziness, headaches
  - Hypotension
  - Bradycardia
- Amlodipine,
- Nifedipine during pregnancy
  - But both CCBs are pregnancy category C



https://pharmacologycorner.com/calcium-channel-blockers

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Calcium channel blockers cause oedema



May be combined with a diuretic, ACE inhibitor or ARB to decrease oedema

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Other antihypertensives during pregnancy

Methyldopa is a pregnancy category A drug

- Indicated in hypertension during pregnancy, gestational pregnancy, pre-eclampsia
  - not generally used in other settings
- It is an alpha 2 agonist*
  - *not an antagonist, remember this receptor is the 'negative feedback receptor' that we put off to the side when talking about sympathomimetics – it does the opposite thing that we would expect a sympathomimetic to do
- Methyldopa causes many psychological side effects and is normally not used outside pregnancy

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Medications of the Cardiovascular System:
Antihypertensive Medications

Lesson 9: Overview of diuretics / Final words about hypertension

(For full diuretics discussion, please refer to the final chapter in this module.)

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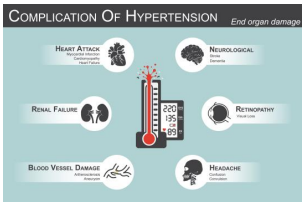
Diuretics

- Inhibit sodium reabsorption from the kidney
- Salt stays in the nephron, and is excreted along with the water that it pulls along with it.
- Some diuretics actually result in compensation, thereby negating the blood pressure lowering effects.
- Thiazide diuretics are a first line therapy for hypertension
- Can be combined with other antihypertensive agents for greater efficacy.

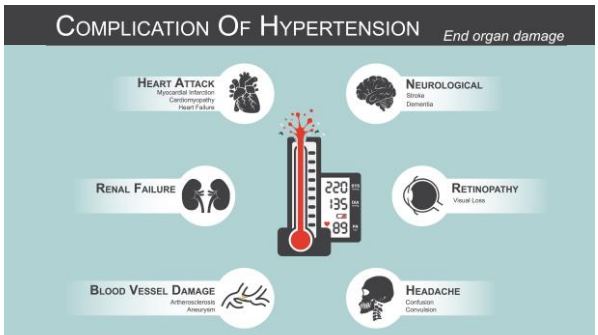
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Final words about treating hypertension

- Initially, diet and lifestyle changes
- Next, one of the first line therapies
  - ACE inhibitors or ARBs
  - Calcium channel blockers
  - Thiazide diuretics
- Then, more than one of the above.



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Thiazide diuretics (hydrochlorothiazide)

- Inhibit sodium reabsorption from the distal tubule
- Wastes potassium as well
- Used for hypertension, congestive heart failure, nephrolithiasis and diabetes insipidus
- Thiazide diuretics compete with the secretion of uric acid so they are contraindicated in people with gout

- Several drugs use hydrochlorothiazide in combination to prevent – a little bit increases the action of other
  - Hydrochlorothiazide (Dithiazide)

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Summary by mode of action

- Decrease heart rate and force
  - Beta blockers
  - Alpha and beta adrenergic blockers
- Decrease vascular resistance
  - Alpha antagonists,
  - ACE inhibitors,
  - Calcium channel blockers,
  - Alpha and beta adrenergic blockers,
  - Angiotensin II receptor blockers
- Decrease plasma volume
  - Angiotensin II receptor blockers
  - Diuretics
  - ACE inhibitors

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